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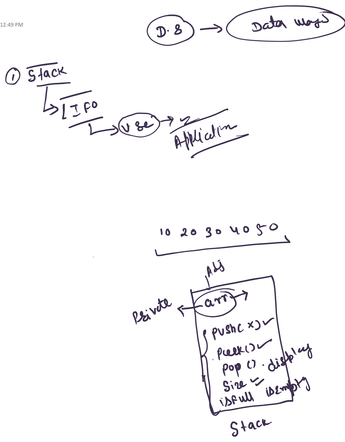
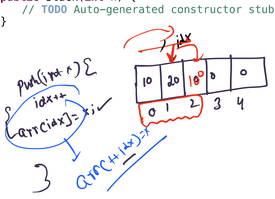
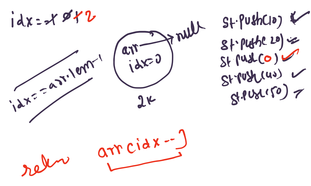
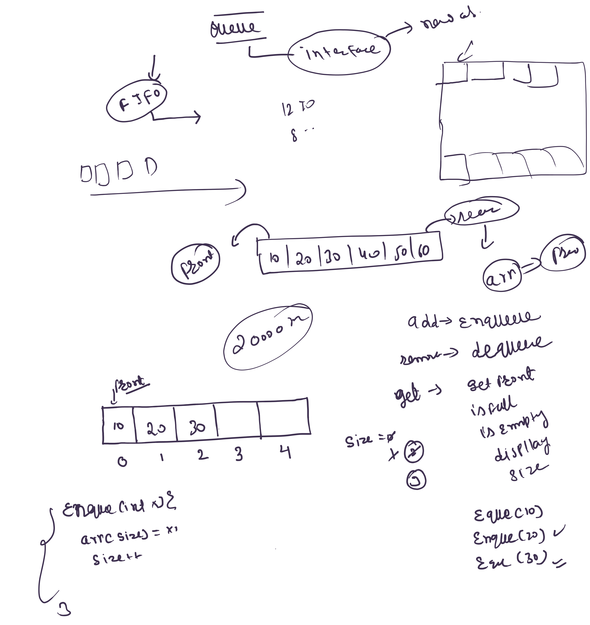



```
public class Stack {
    private int[] arr;
    private int idx;

    public Stack() {
        // TODO Auto-generated constructor stub
    }
    public Stack(int n) {
        // TODO Auto-generated constructor stub
    }
}
```

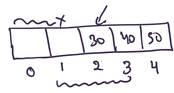
```
public class Stack_Client {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        Stack st = new Stack(100);
    }
}
```





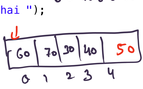


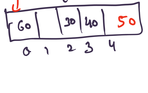

```
public boolean isEmpty() {
        return size == 0;
    }

    public boolean isFull() {
        return size == arr.length;
    }

    public int size() {
        return size;
    }

    public void Enqueue(int x) throws Exception {
        if (isFull()) {
            throw new Exception("Bklol Queue full hai ");
        }
        arr[size] = x;
        size++;
    }
```




```
    public int  Dequeue() throws Exception {
            if(isEmpty()) {
                throw new Exception("Bklol Queue Empty hai ");
            }
            int x=arr[front];
            front ++;
             size--;
             return x;
          }
```

```
public void Enqueue(int x) throws Exception {
        if (isFull()) {
            throw new Exception("Bklol Queue full hai ");
        }
        int idx = front + size;
        arr[idx] = x;
        size++;
    }

    public int Dequeue() throws Exception {
        if (isEmpty()) {
            throw new Exception("Bklol Queue Empty hai ");
        }
        int x = arr[front];
        front++;
        size--;
        return x;
    }

    public int getFront() throws Exception {
        if (isEmpty()) {
            throw new Exception("Bklol Queue Empty hai ");
        }
        return arr[front];
    }
```





Size = 4

i=0 f+0 = 2+0 = 2 → 30
i=1 f+1 = 2+1 = 3 → 40
i=2 f+2 = 2+2 = 4 → 50
i=3 f+3 = 2+3 = 5 % 5 = 0 → 60

```
for(i=0 ; i<size ; i++)
int idx=(f+i)%arr.len
Sys o arr[idx]
```